

Algebraic Identities — Practice

Section A is interactive. Sections B and C are written practice — check them against the separate answer key.

Objective (1 mark each)

Q1 • MCQ**[1]**

$(a + b)^2$ equals:

(a) $a^2 + b^2$ (b) $a^2 + 2ab + b^2$ (c) $a^2 - 2ab + b^2$ (d) $2a + 2b$

Q2 • MCQ**[1]**

$(a + b)(a - b)$ equals:

(a) $a^2 + b^2$ (b) $a^2 - b^2$ (c) $a^2 - 2ab + b^2$ (d) $2ab$

Q3 • FILL IN THE BLANK**[1]**

$(a - b)^2 = a^2 - \underline{\hspace{2cm}} + b^2$. (Fill the middle term.)

Q4 • MCQ**[1]**

Using an identity, 99^2 equals:

(a) 9081 (b) 9801 (c) 9999 (d) 8901

Q5 • TRUE / FALSE**[1]**

$(a - b)^2$ and $(b - a)^2$ are always equal.

True False

Q6 · ASSERTION-REASON**[1]****Assertion (A):** $397 \times 403 = 159991$.**Reason (R):** $(a + b)(a - b) = a^2 - b^2$.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q7**[2]**Expand using an identity: (i) $(m + 3)^2$ (ii) $(6 + p)^2$.

Q8**[2]**Expand $(6x + 5)^2$.

Q9**[2]**Find 55^2 using a suitable identity.

Long answer (3 marks each)

Q10**[3]**Use identities to compute: (i) 46^2 (ii) 397×403 .

Q11

[3]

Expand: (i) $(3j + 2k)^2$ (ii) $(2x - 5y)^2$.

Q12

[3]

Without fully multiplying, decide which is larger: 14×26 or 16×24 . Justify using the identity $(a - b)(a + b) = a^2 - b^2$.

Total: 21 marks · Check your work against the answer key.